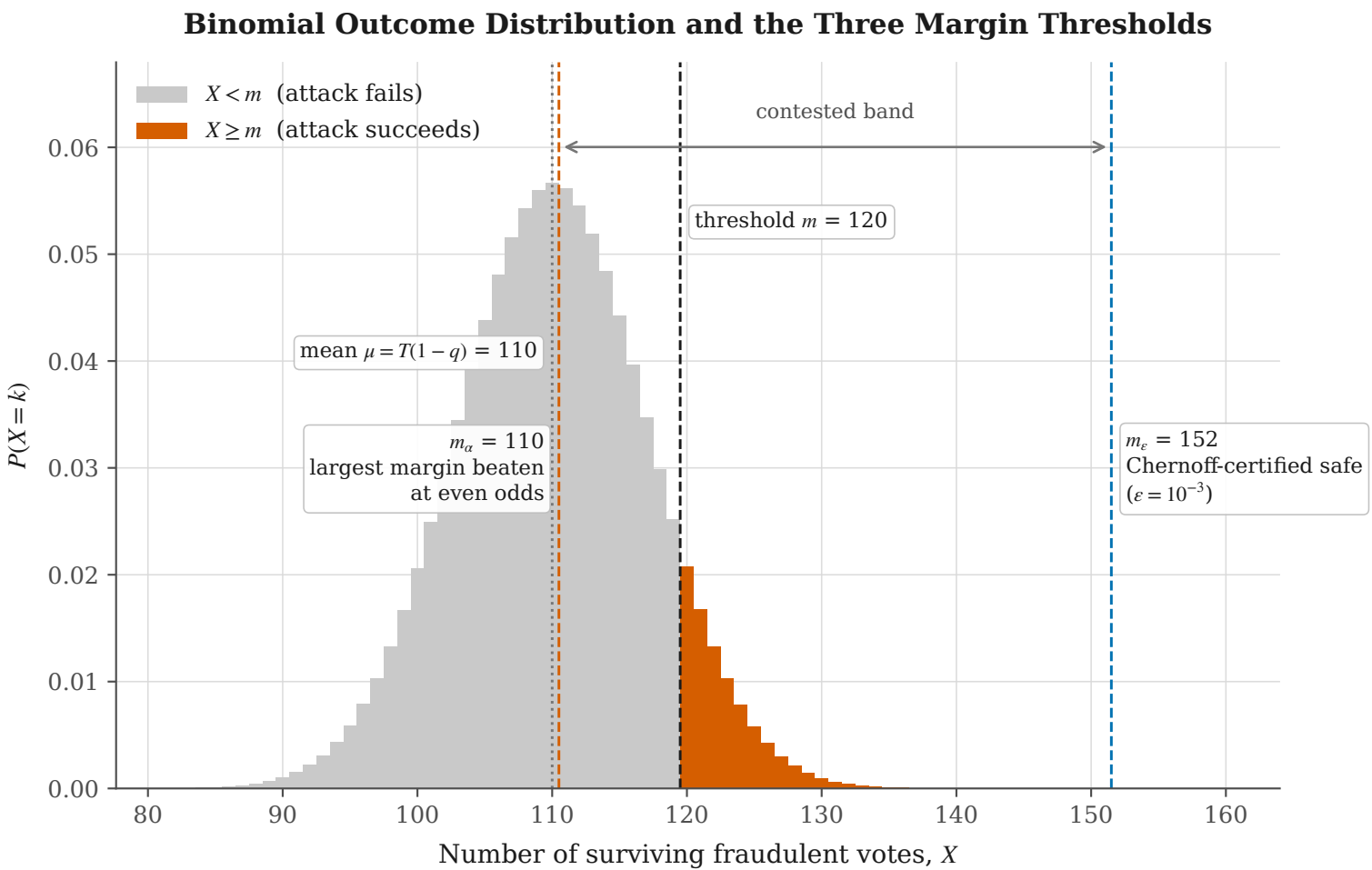

$X \sim \text{Binomial}(T = 200,\ 1 - q = 0.55)$; the shaded success region has probability $P(X \geq m) = 0.088$. Because X is integer-valued, region boundaries are drawn at half-integers (the m line sits at $m - 0.5$), so each line meets the edge of a bar rather than cutting through its centre. Margins at or below m_α are comfortably within reach; margins at or above m_ε are provably out of reach.